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Mathematical Induction: Sum Formulas & Proof by Induction

Student: Justin Koo **Date:** 2026-05-10

Standard: *CCSS.MATH.CONTENT.HSA-APR — Use mathematical induction to prove statements about sums, divisibility, and inequalities; write rigorous base case + inductive step arguments.*

Proof by Mathematical Induction — Two-Step Framework

Step 1 — Base Case: Show the statement $P(n)$ is true for $n = 1$ (or the first relevant integer).

Step 2 — Inductive Step: Assume $P(k)$ is true (inductive hypothesis). Then *prove* $P(k+1)$ is true.

Conclusion: By the Principle of Mathematical Induction, $P(n)$ is true for all $n \geq 1$.

Problem 1 — Sum of First n Positive Integers

Difficulty: Introductory. *Follow the template below. Write each step clearly.*

Worked Example — Sum of First n Odd Numbers

Prove by induction: $1 + 3 + 5 + \cdots + (2n-1) = n^2$ for all $n \geq 1$.

Base Case ($n = 1$): LHS $= 1$. RHS $= 1^2 = 1$. ✓

Inductive Hypothesis: Assume $1 + 3 + \cdots + (2k-1) = k^2$ for some $k \geq 1$.

Inductive Step (prove for $n = k+1$):

$$\begin{aligned} &1 + 3 + \cdots + (2k-1) + (2(k+1)-1) \\ &= k^2 + (2k+1) \quad \text{(by inductive hypothesis)} \\ &= (k+1)^2 \quad \text{(factor)} \end{aligned}$$

Conclusion: The statement holds for all $n \geq 1$. ■

1. Prove by mathematical induction that for all integers $n \geq 1$:

$$1 + 2 + 3 + \cdots + n \;=\; \frac{n(n+1)}{2}$$

Base Case ($n = 1$):

Inductive Hypothesis:

Assume the formula holds for $n = k$, i.e., $\sum_{i=1}^{k} i =$ _____

Inductive Step: Show it holds for $n = k + 1$.

Conclusion:

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Problem 2 — Sum of Squares

Difficulty: Easy. *The algebra in the inductive step requires careful factoring. Show every line.*

Worked Example — Sum of First n Even Numbers

Prove by induction: $2 + 4 + 6 + \cdots + 2n = n(n+1)$ for all $n \geq 1$.

Base Case ($n = 1$): LHS $= 2$. RHS $= 1 \cdot 2 = 2$. ✓

Inductive Hypothesis: Assume $2 + 4 + \cdots + 2k = k(k+1)$.

Inductive Step:

$$\begin{aligned} &2 + 4 + \cdots + 2k + 2(k+1) \\ &= k(k+1) + 2(k+1) \quad \text{(by inductive hypothesis)} \\ &= (k+1)(k+2) \quad \text{(factor out } (k+1)\text{)} \\ &= (k+1)\big((k+1)+1\big) \quad \checkmark \end{aligned}$$

Conclusion: $P(n)$ holds for all $n \geq 1$. ■

2. Prove by mathematical induction that for all integers $n \geq 1$:

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

Base Case ($n = 1$):

Inductive Hypothesis:

Inductive Step:

Conclusion:

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Problem 3 — Geometric Series Sum

Difficulty: Medium. *The geometric series formula connects directly to the convergence topics from last week. Watch the algebra when multiplying both sides.*

Worked Example — Powers of 2

Prove by induction: $1 + 2 + 4 + \cdots + 2^{n-1} = 2^n - 1$ for all $n \geq 1$.

Base Case ($n = 1$): LHS $= 2^0 = 1$. RHS $= 2^1 - 1 = 1$. ✓

Inductive Hypothesis: Assume $1 + 2 + \cdots + 2^{k-1} = 2^k - 1$.

Inductive Step:

$$\begin{aligned} &1 + 2 + \cdots + 2^{k-1} + 2^k \\ &= (2^k - 1) + 2^k \quad \text{(by inductive hypothesis)} \\ &= 2 \cdot 2^k - 1 = 2^{k+1} - 1 \quad \checkmark \end{aligned}$$

Conclusion: $P(n)$ holds for all $n \geq 1$. ■

3. Prove by mathematical induction that for all integers $n \geq 1$ and $r \neq 1$:

$$1 + r + r^2 + \cdots + r^{n-1} = \frac{r^n - 1}{r - 1}$$

(Hint: in the inductive step, add r^k to both sides and factor.)

Base Case ($n = 1$):

Inductive Hypothesis:

Inductive Step:

Conclusion:

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Problem 4 — Divisibility Proof

Difficulty: Medium-Hard. *Divisibility proofs use a different structure than sum proofs. The key move is writing $f(k+1)$ in terms of $f(k)$.*

Worked Example — Divisibility by 3

Prove by induction: $n^3 - n$ is divisible by 3 for all $n \geq 1$.

Base Case ($n = 1$): $1^3 - 1 = 0 = 3 \cdot 0$. Divisible by 3. ✓

Inductive Hypothesis: Assume $k^3 - k = 3m$ for some integer m.

Inductive Step:

$$\begin{aligned}
&(k+1)^3 - (k+1) \\
&= k^3 + 3k^2 + 3k + 1 - k - 1 \\
&= (k^3 - k) + 3k^2 + 3k \\
&= 3m + 3(k^2 + k) \quad \text{(by inductive hypothesis)} \\
&= 3(m + k^2 + k), \text{ which is divisible by 3. } \checkmark
\end{aligned}$$

Conclusion: $3 \mid (n^3 - n)$ **for all** $n \geq 1$. ■

4. Prove by mathematical induction that $6 \mid (n^3 - n)$ for all integers $n \geq 1$.

That is, prove that $n^3 - n$ is always divisible by 6.
(Hint: factor $n^3 - n = n(n-1)(n+1)$ first. In the inductive step, write $(k+1)^3 - (k+1)$ and separate it into $f(k)+$ extra terms.)

Base Case ($n = 1$):

Inductive Hypothesis:

Inductive Step:

Conclusion:

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Problem 5 — Inequality Proof

Difficulty: Hard. *Inequality induction requires careful reasoning about the direction of the inequality at each step. Justify every inequality sign.*

Worked Example — Exponential vs. Linear

Prove by induction: $2^n > n$ for all $n \geq 1$.

Base Case ($n = 1$): $2^1 = 2 > 1$. ✓

Inductive Hypothesis: Assume $2^k > k$ for some $k \geq 1$.

Inductive Step:

$2^{k+1} = 2 \cdot 2^k > 2k$ (by inductive hypothesis, multiply both sides by 2)

$2k = k + k \geq k + 1$ (since $k \geq 1$)

Therefore $2^{k+1} > k + 1$. ✓

Conclusion: $2^n > n$ for all $n \geq 1$. ■

5. Prove by mathematical induction that for all integers $n \geq 4$:

$$2^n \geq n^2$$

(Note: the base case starts at $n = 4$, not $n = 1$. Check $n = 1, 2, 3$ to see why they fail.)
(Hint for inductive step: show $2^{k+1} = 2 \cdot 2^k \geq 2k^2$, then compare $2k^2$ with $(k+1)^2 = k^2 + 2k + 1$.)

Base Case ($n = 4$):

Inductive Hypothesis:

Inductive Step:

Conclusion: